Unit 02: Operating Systems

CONTENTS

Objectives

After studying this unit, you will be able to:

- Basics of the operating system.
- Purposes of the operating system.
- Types of the operating system.
- Explain the purpose of the operating system.
- Discuss the types of operating systems.
- Explain the user interfaces.

Introduction

Modern multiprocessing OSs allow many processes to be active, where each process is a "thread" of computation being used to execute a program. One form of multiprocessing is called time-sharing, which lets many users share computer access by rapidly switching between them. Time-sharing must guard against interference between users' programs, and most systems use virtual memory, in which the memory, or "address space," used by a program may reside in secondary memory (such as on a magnetic hard disk drive) when not in immediate use, to be swapped back to occupy the faster main computer memory on demand. This virtual memory both increases the address space available to a program and helps to prevent programs from interfering with each other, but it requires careful control by the operating system and a set of allocation tables to keep track of memory use. Perhaps the most delicate and critical task for a modern OS is allocation of the CPU; each process is allowed to use the CPU for a limited time, which may be a fraction of a second, and then must give up control and become suspended until its next turn. Switching between processes must use the CPU while protecting all data of the processes.

2.1 Operating System

An Operating System (OS) is an important part of almost every computer system. A collection of programs that manage and coordinate the activities taking place within a computer. An OS is specialized software that controls and monitors the execution of all other programs that reside in the computer, including application programs and other system software. It aids in the execution of the user programs and makes solving the user problems easier. An OS acts as an interface between the software and the computer hardware. It is an intermediary between the user of a computer and the computer hardware and application programs

Figure 1).

Additionally, the **OS** manages a computer's resources, especially the allocation of those resources among other programs. Typical resources include the Central Processing Unit (CPU), computer memory, file storage, input/output (I/O) devices, and network connections. Management tasks include scheduling resource use to avoid conflicts and interference between programs. Unlike most programs, which complete a task and terminate, an operating system runs indefinitely and terminates only when the computer is turned off.

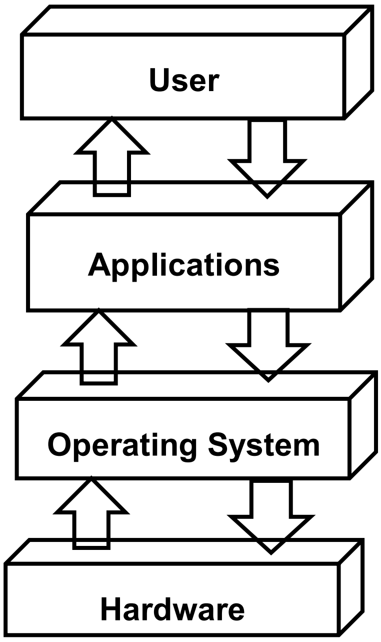

Figure 1: Operating System as an Interface

The first digital computers had no OSs. They ran one program at a time, which had command of all system resources, and a human operator would provide any special resources needed. The first OSs were developed in the mid-1950s. These were small "supervisor programs" that provided basic I/O operations (such as controlling punch card readers and printers) and kept accounts of CPU usage for billing. The supervisor programs also provided multiprogramming capabilities to enable several programs to run at once. This was particularly important so that these early multimillion-dollar machines would not be idle during slow I/O operations.

A computer system can be divided roughly into four components—the hardware, the operating system, the application programs, and the users (Figure 2). The hardware-the CPU, the memory, and the I/O devices provide the basic computing resources. The application programs, such as word processors, spreadsheets, compilers, and web browsers-define how these resources are used to solve the computing problems of the users. The OS controls and coordinates the use of the hardware among the various application programs for the various users. The OS provides the means for the proper use of these resources in the operation of the computer system. An operating system is similar to a government. Like a government, it performs no useful function by itself. It simply provides an environment within which other programs can do useful work.

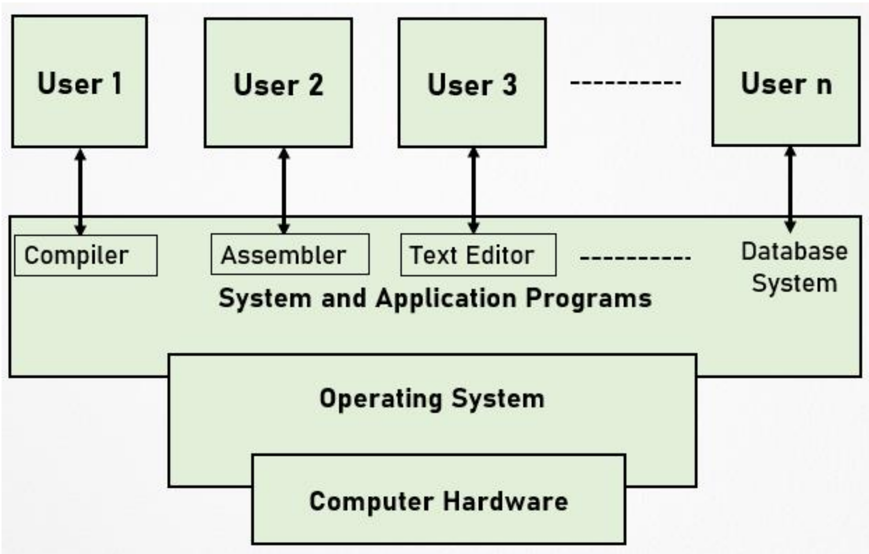

Figure 2: Abstract View of the Components of a Computer System

Objectives of Operating Systems

An OS is an integrated set of specialized programs used to manage the overall resources and operations of the computer. It helps in making the computer system convenient to use (adding *convenience*). The main objective of an OS is to manage all the other resources and operations of the computer system. It helps to make computing much more convenient much easy to use. The problem-solving capabilities can be more easily done with the help of an OS.

Another important aspect when we talk about OS is the capability of abstraction. You must have studied out in different programming languages of the computer that there is a term called abstraction. Abstraction deals out to hide the inner details from the users. So, you just get an illusion of a computer system that is completely dedicated to you but what is going in around into it. You are not concerned with that you just have to input your data, and you get output but how it is working, who is controlling that operation, and what is the logic that is being controlled out how memory is accessed, how processors are managed, how the programs are executed. All these operations are carried out by an OS. It acts as an intermediary between the hardware and its users, making it easier for the users to access and use other resources facilitating the use of the computer hardware efficiently.

The OS provides the users with an interactive interface to use the computer system/ *Interfacing with the Users* (typically via a GUI). It provides an interactive interface that is one to one interface between the user and the computer system. The computer system interfacing with the users is much easier, this is facilitated with the help of a Graphical User Interface (GUI).

Then, it helps out to keep track of who is using which resource, how the resources are being scheduled out. It helps in granting the resource requests. Suppose one user wants access to any particular file on a system. The file is also a resource one user wants to access the printer. Here, the printer is also a resource. If a user wants to perform some calculation, here, the processor is also a resource. The decisions related to how resources are allocated, how grant to those resources is done are done by an OS. It helps in mediating any kind of conflicts also so if one user requests a printer, another user also requests a printer. The printer is required by both the users. An OS ensures efficient and fair sharing of resources, among the users, as well as the user programs.

2.2 Functions of an Operating System

The OS performs a large variety of functions that include:

Booting the Computer: The first most important function of an OS is booting. It helps in loading the essential part of an OS, that is, the Kernel which is the most basic level or core of an OS. The kernel performs all the tasks related to booting, resource allocation, file management, security by an OS. The booting of the computer system also involves reading the opening batch of instructions for a computer system to start. Initially, the kernel is loaded out into the computer system, and to the memory main memory of the computer system. It also helps to determine the hardware connected to a computer.

Configuring Devices: The connecting of different kinds of hardware resources determining which resources are connected to a computer system involves the configuration of devices that are connected to it. Different device drivers are needed out to run the devices. Such device drivers can be reinstalled or can be updated. The connection of the various USB to the computer system is also determined by an OS. The concept of plug-and-play of the devices you connect to the computer system is extensively done and recognized automatically by an OS.

Managing Network Connections: The management of network connections is done by an OS. It helps to manage the wired connections to a computer system as well as detecting the wireless connections at your home, your school or work, or wherever you go. The remote connections are also maintained out through the help of this OS. Usually, the wired and wireless adapters are all controlled out by an OS.

Managing and Monitoring Resources and Jobs: Another objective of an OS is management and monitoring of resources and jobs. This involves the management of different resources. If a user wants to access any particular file, another user wants to access a printer, the third user wants to access, another device in a computer system. All requests to these resources are also provisioned out with the help of an OS. In case a deadlock arises, like, both, there are two users or three users. All of these want to access a printer, so, in that situation to which user a printer is required to be given, and for how much time it is required to be given that is all decided out with the OS. The management of resources, availability of resources making the resources available to any user any program or any devices, this is all done by an OS. The monitoring tracks the usage and duration of usage of the resources. The pending requests for accessing the resources and their priorities are decided by an OS. The additional monitoring activities or tracking activities are also done by an OS.

Job Scheduling: An OS performs the task of scheduling the jobs. Suppose you have a printer with a scheduled job of printing five different files. You give print commands for all those files so that job scheduling, file 1, file 2, file 3, file 4, and file 5 are waiting for printing. All these are managed or scheduled in a particular order as decided by an OS.

Security: An OS prevents unauthorized access to programs and data using passwords and other similar techniques. It offers password protection; monitoring the biometric characteristics and providing biometric access; offering security through firewalls.

Core Functions of an Operating System

Memory Management: Main memory is a large array of words or bytes where each word or byte has its address. The main memory provides fast storage that can be accessed directly by the CPU. For a program to be executed, it must in the main memory. The OS manages the main memory. It keeps tracks of primary memory, that is, what art of it are in use by whom, what part are not in use. The OS decides which process will get memory when and how much. It allocates the memory when a process requests it to do so as well as deallocates the memory when a process no longer needs it or has been terminated.

Processor Management: In a multiprogramming environment, the OS decides which process gets the processor when and for how much time. This function is called process scheduling. An OS acts as processor management. It does the following activities for processor management −

- Keeps track of the processor and status of the process.
- The program responsible for tracking the processor and its status is known as the traffic controller.
- Allocates the processor (CPU) to a process.
- De-allocates processor when a process is no longer required.

Device Management: An OS manages the device communication via their respective drivers. The following activities for device management are carried out by OS:

- Keeps tracks of all devices.
- The program responsible for this task is known as the I/O controller.

- Decides which process gets the device when and for how much time.
- Allocates the device in an efficient way.
- De-allocates devices.

File Management: The files in a computer system are viewed out in a hierarchical format. There is a hierarchy that is established, the files can be organized out into directories and subdirectories can be there, which helps in easy navigation and usage so all the records. An OS manages the files and helps to keep track of the information that is stored in a file. The location of a file, who are the users of it; when the file was generated out; when it was edited out; and what is the status of a file are all managed and tracked by an OS with the help of maintaining a file system.

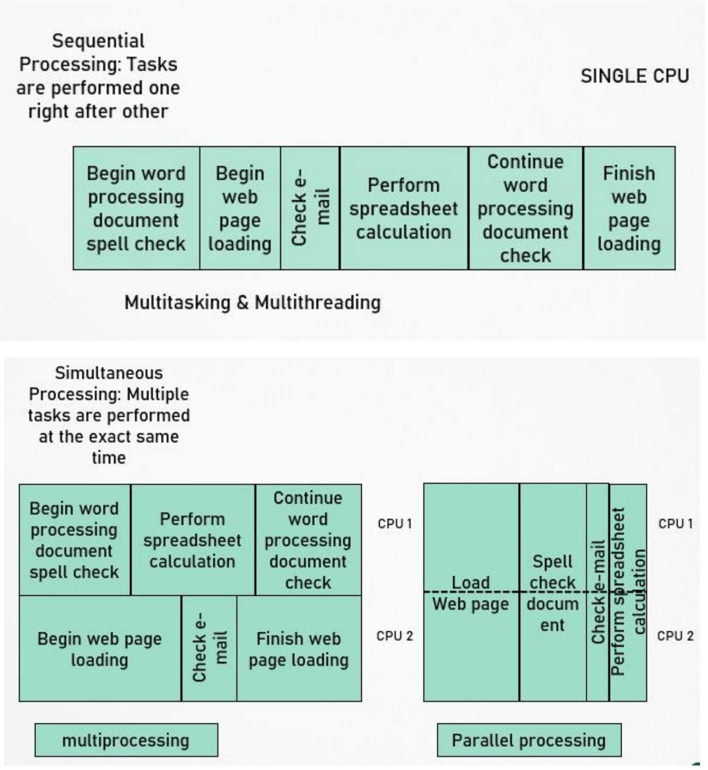

Figure 3: Sequential vs Multiprocessing

Additional OS Responsibilities

- **Job Accounting**– Keeps track of time and resources used by various jobs and/or users.
- **Control Over System Performance**– Records delays between the request for a service and from the system.
- **Interaction with the Operators**– Interaction may take place via the console of the computer in the form of instructions. The OS acknowledges the same, does the corresponding action and informs the operation by a display screen.
- **Error-detecting Aids**– Production of dumps, traces, error messages, and other debugging and error-detecting methods.
- **Coordination Between Other Software and Users** – Coordination and assignment of compilers, interpreters, assemblers, and other software to the various users of the computer systems.
- **Multitasking and Multithreading:** The ability of an OS to have more than one program (task) open at one time is called Multitasking. OS enables a CPU on rotating between tasks doing the switching. The ability to rotate between multiple threads so that processing is completed faster and more efficiently. An OS monitors that the sequence of instructions within a program is independent of other threads.
- **Multiprocessing and Parallel Processing:** Multiple processors (or multiple cores) are used in one computer system to perform work more efficiently. Multiprocessing is handled by an OS. Figure 3 depicts the sequential and multiprocessing concept.

2.3 Operating System Kernel

The Kernel is the central component of most computer OSs that acts as a bridge between the applications and the actual data processing done at the hardware level. It is also called the nucleus or core and was conceived as containing only the essential support features of the OS (Figure 4). The kernel is responsible for managing the system's resources (such as communication between the hardware and the software components) and can provide the lowest-level abstraction layer for the resources (especially processors and I/O devices). It decides which memory each process can use, and determining what to do when not enough memory is available. It makes the facilities available to application processes through inter-process communication mechanisms and system calls.

The kernel is the most significant OS component that decides at any time which of the many running programs should be allocated to the processor or processors. It allocates requests from applications to perform I/O to an appropriate device. In a monolithic kernel, all OS services run along with the main kernel thread, thus also residing in the same memory area. In the microkernel approach, the kernel itself only provides basic functionality that allows the execution of servers, separate programs that assume former kernel functions, such as device drivers, GUI servers, etc.

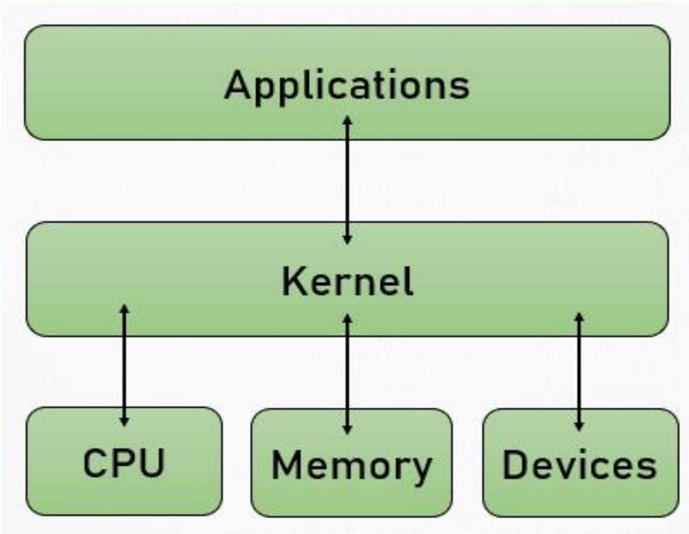

Figure 4: Kernel Interface with Applications and Computer Components

System Call Model in Operating System

A system call is a mechanism used by an application to request service from the OS based on the monolithic kernel or to system services on the OSs based on the microkernel structure. System Calls are requests by programs for services to be provided by the OS.

System call represents the interface of a program to the kernel of the OS with the following characteristics:

- The system call API (Application Program Interface) is given in C.
- The system call is a wrapper routine typically consisting of a short piece of assembly language code.
- The system call generates a hardware trap and passes the user's arguments and an index into the kernel's system call table for the service requested.
- The kernel processes the request by looking up the index passed to see the service to perform and carries out the request.
- Upon completion, the kernel returns a value that represents either successful completion of the requestor an error. If an error occurs, the kernel sets a global variable, err no, indicating the reason for the error.
- The process is delivered the return value and continues its execution.

2.4 Types of Operating Systems

Within the broad family of operating systems, there are different types, categorized based on the types of computers they control and the sort of applications they support (Figure 5).

Batch Processing

This type of operating system does not interact with the computer directly. There is an operator which takes similar jobs having the same requirement and groups them into batches (Figure 6). It is the responsibility of the operator to sort jobs with similar needs. Some computer processes are very lengthy and time-consuming. To speed the same process, a job with a similar type of needs is batched together and run as a group. The user of a batch operating system never directly interacts with the computer. In this type of OS, every user prepares his or her job on an offline device like a punch card and submits it to the computer operator. Examples of the batch OS are the Payroll system, Bank statements, etc.

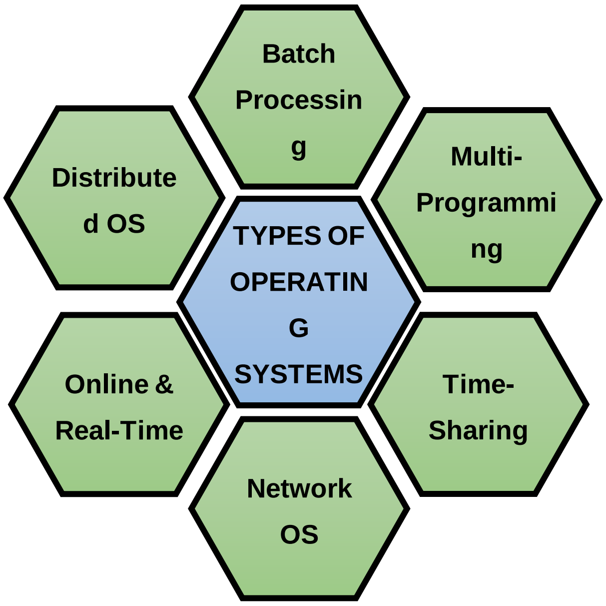

Figure 5: Different Types of Operating Systems

Advantages of Batch Operating System:

- It is very difficult to guess or know the time required for any job to complete. Processors of the batch systems know how long the job would be when it is in queue
- Multiple users can share the batch systems
- The idle time for the batch system is very less
- It is easy to manage large work repeatedly in batch systems

Disadvantages of Batch Operating System:

- The computer operators should be well known with batch systems
- Batch systems are hard to debug
- It is sometimes costly
- The other jobs will have to wait for an unknown time if any job fails

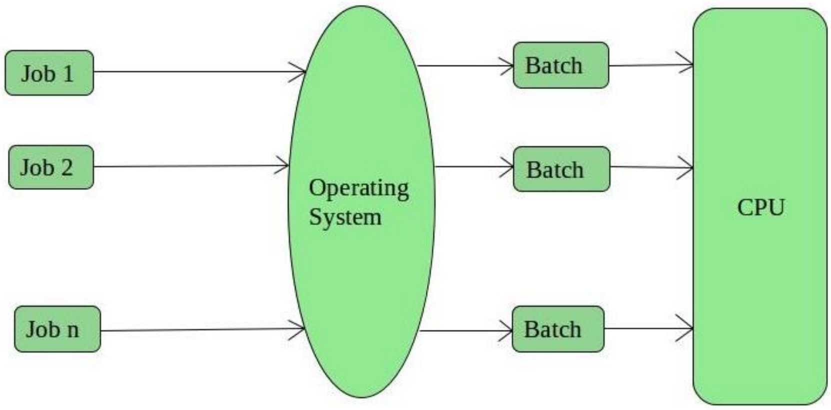

Figure 6: Batch Operating System Environment

Multi-Programming Operating System

In the multi-programming system, one or multiple programs can be loaded into its main memory for execution. Only one program or process is capable to get CPU for the execution of the instructions while other programs wait for getting their turn. Primarily, the objective of multiprogramming is to manage entire resources of the system that includes system components such as command **processor**, file system, I/O control system, and transient area. A multiprogramming system helps to overcome the issue of under-utilization of CPU & primary memory.

Time-Sharing Operating System

Each task is given some time to execute so that all the tasks work smoothly. Each user gets the time of CPU as they use a single system. These systems are also known as multitasking systems. The task can be from a single user or different users also. The time that each task gets to execute is called quantum. After this time interval is over OS switches over to the next task.

Advantages of Time-Sharing OS:

- Each task gets an equal opportunity
- Fewer chances of duplication of software
- CPU idle time can be reduced

Disadvantages of Time-Sharing OS:

- Reliability problem
- One must have to take care of the security and integrity of user programs and data
- Data communication problem

Network Operating System (NOS)

These systems run on a server and provide the capability to manage data, users, groups, security, applications, and other networking functions. These types of OSs allow shared access of files, printers, security, applications, and other networking functions over a small private network. One more important aspect of NOS is that all the users are well aware of the underlying configuration, of all other users within the network, their connections, etc. and that's why these computers are popularly known as tightly coupled systems.

Advantages of Network Operating System:

- Highly stable centralized servers that handle the security concerns
- New technologies and hardware up-gradation are easily integrated into the system
- Server access is possible remotely from different locations and types of systems

Disadvantages of Network Operating System:

- Servers are costly
- User has to depend on a central location for most operations
- Maintenance and updates are required regularly

Real-Time Operating System (RTOS)

Real-time operating systems are used to control machinery, scientific instruments, and industrial systems such as embedded systems (programmable thermostats, household appliance controllers), industrial robots, spacecraft, industrial control (manufacturing, production, power generation, fabrication, and refining), and scientific research equipment. An RTOS typically has the very little user interface capability, and no end-user utilities, since the system will be a "sealed box" when delivered for use. A very important part of an RTOS is managing the resources of the computer so that a particular operation executes in precisely the same amount of time, every time it occurs. In a complex machine, having a part move more quickly just because system resources are available may be just as catastrophic as having it not move at all because the system is busy.

An RTOS facilitates the creation of a real-time system, but does not guarantee the final result will be real-time; this requires correct development of the software. An RTOS does not necessarily have high throughput; rather, an RTOS provides facilities that, if used properly, guarantee deadlines can be met generally (soft real-time) or deterministically (hard real-time). An RTOS will typically use specialized scheduling algorithms to provide the real-time developer with the tools necessary to produce deterministic behavior in the final system. An RTOS is valued more for how quickly and/or predictably it can respond to a particular event than for the given amount of work it can perform over time.

Distributed Operating System

It is a recent advancement in the world of computer technology and is being widely accepted all over the world and, that too, with a great pace. Various autonomous interconnected computers communicate with each other using a shared communication network. Independent systems possess their memory unit and CPU. These are referred to as loosely coupled systems or distributed systems. These system's processors differ in size and function. The major benefit of working with these types of the operating system is that it is always possible that one user can access the files or software which are not present on his system but some other system connected within this network i.e., remote access is enabled within the devices connected in that network.

Advantages of Distributed Operating System:

- Failure of one will not affect the other network communication, as all systems are independent of each other
- Electronic mail increases the data exchange speed
- Since resources are being shared, computation is highly fast and durable
- Load on host computer reduces
- These systems are easily scalable as many systems can be easily added to the network
- Delay in data processing reduces

Disadvantages of Distributed Operating System:

- Failure of the main network will stop the entire communication
- To establish distributed systems the language which is used are not well defined yet
- These types of systems are not readily available as they are very expensive. Not only that the underlying software is highly complex and not understood well yet

Task: Explore the different types of Operating Systems along with their characteristics.

2.5 Providing a User Interface

The user interface is the space where interaction between humans and machines occurs. The goal of interaction between a human and a machine at the user interface is effective operation and control of the machine, and feedback from the machine which aids the operator in making operational decisions. A user interface is a system by which people (users) interact with a machine. The user interface includes hardware (physical) and software (logical) components. User interfaces exist for various systems and provide a means of:

- Input, allowing the users to manipulate a system, and/or
- Output, allowing the system to indicate the effects of the users' manipulation.

The following types of the user interface are the most common:

1. **Graphical User Interfaces (GUIs):** Most modern operating systems, like Windows and the Macintosh OS, provide a graphical user interface (GUI). A GUI lets you control the system by using a mouse to click *graphical objects* on the screen. A GUI is based on the desktop metaphor. Graphical objects appear on a background (the desktop), representing resources you can use. *Icons* are pictures that represent computer resources, such as printers, documents, and programs. You double-click an icon to choose (activate) it, for instance, to launch a program. The Windows operating system offers two unique tools, called the taskbar and the Start button. These help you run and manage programs.
2. **Command-Line Interfaces:** A CLI (command line interface) is a user interface to a computer's operating system or an application in which the user responds to a visual prompt by typing in a command on a specified line, receives a response from the system, and then enters another command, and so forth. The MS-DOS Prompt application in a Windows operating system is an example of the provision of a command-line interface. Today, most users prefer the graphical user interface (GUI) offered by Windows, Mac OS, BeOS, and others. Typically, most of today's UNIX-based systems offer both a command-line interface and a graphical user interface.

Case Study: In the 1980s UNIX, VMS and many others had operating systems that were built this way. GNU/Linux and Mac OS X are also built this way. Modern releases of Microsoft Windows such as Windows Vista implement a graphics subsystem that is mostly in user-space; however, the graphics drawing routines of versions between Windows NT 4.0 and Windows Server 2003 exist mostly in kernel space. Windows 9x had very little distinction between the interface and the kernel.

2.6 Running Programs

One of the most important X features is that windows can come either from programs running on another computer or from an operating system other than UNIX. So, if your favorite MS-DOS program doesn't run under UNIX but has an X interface, you can run the program under MSDOS and display its windows with X on your UNIX computer. Researchers can run graphical data analysis programs on supercomputers in other parts of the country and see the results in their offices.

Setting Focus

Of all the windows on your screen, only one window receives the keystrokes you type. This window is usually highlighted in some way. By default, in ***the mwm window manager,*** for instance, the frame of the window that receives your input is a darker shade of grey. In X jargon, choosing the window you type to is called "setting the *input focus*." Most window managers can be configured to set the focus in one of the following two ways:

(*a*) Point to the window and click a mouse button (usually the first button). You may need to click on the title bar at the top of the window.

(*b*) Simply move the pointer inside the window. When you use **mwm,** any new windows will get the input focus automatically as they pop up.

The Xterm Window

One of the most important windows is an xterm window. The xterm makes a terminal emulator window with a UNIX login session inside, just like a miniature terminal. You can also start separate X-based window programs (typically called clients) by entering commands in an xterm window.

The Root Menu

If you move the pointer onto the root window (the "desktop" behind the windows) and press the correct mouse button (usually the first or third button, depending on your setup), you should see the root menu. It has commands for controlling windows. The menu's commands may differ depending on the system. The system administrator (or window manager) can add commands to the root menu. These can be window manager operations or commands to open other windows.

2.7 Sharing Information

UNIX makes it easy for users to share files and directories. It involves controlling exactly who has access takes some explaining. The directory access permissions exist that help to control access to the files in it. It affects the overall ability to use files and subdirectories in the directory.

There is a different kind of file access permissions, any kind of file access from any kind of access permissions that are used for controlling the content of a particular file, who will be able to access it, which user will be able to edit it which user will be able to just view it, that is all included out into this file access permissions. These access permissions support controlling a particular file content. With the access permissions on a particular directory as well where the file is kept under control whether the file can be renamed or removed or whether it can be edited or not.

Certain files can be made private files. In case you want to make a particular file private, only you can edit it. We use ch mod 600 commands as a terminal emulator. To provide access to the user for editing, we can use a command ch mod 600 filenames. To protect a particular file from any kind of accidental editing in cases suppose a user starts editing it and it is not provided out for editing, command ch mod 400 file name is used. In order to edit a particular file and let everyone else on a particular system only read it without editing, then we use ch mod 644 filename. Now, in case you want to provide that particular file to the non-group users to read it, but not edit it then we use the commands, we use ch mod 664 filename. In case you want to specify any particular file to be read and edited by anyone else, then, there we use a CH mod 666 file name. We have different kinds of file and directory, sub-directory permissions that are provided out to an OS using different kinds of commands.

2.8 Managing Hardware

It is possible to supply command-line options when starting the Hardware Management Agent manually. Use the command-line options as follows:

hwagentd *OPTIONS*

The command line options are explained in the following Table 1.

Table 1: Command-Line Options

Options	Function
-h	Displays the usage message and exits
-v	Displays the hardware Management Agent version currently installed on this Sun x 64 Server and exits
-l level	Override the logging level set in the hwagentd.conf file with level

When using the logging levels option, you must supply a decimal number to set the logging level to use. This decimal number is calculated from a bitfield, depending on the logging level you want to specify. For more information on the bit field used to configure different log levels.

Hardware Management Agent Configuration File

Once the Hardware Management Agent and Hardware SNMP Plugins are installed on the Sun x64 server. There is only one configuration file for the Hardware Management Agent, which configures the level of detail used for log messages. It records log messages into the log file. These messages can be used to troubleshoot the running status of the Hardware Management Agent. Once the Hardware Management Agent and Hardware SNMP Plugins are installed on the Oracle server you want to monitor, you can configure the level of detail used for log messages using the hwmgmtd.conf file.

Configuring the Hardware Management Agent Logging Level

To configure the logging level, modify the *hwagentd_log_levels* parameter in the hwagentd. conf file. The *hwagentd_log_levels* parameter is a bit flag set expressed as a decimal integer. In order to configure the Hardware Management Agent Logging Level, the following steps are followed:

a) Depending on the host OS, the Hardware Management Agent is running on, open the hwagentd.conf file. You can use any text editor to modify this file.

b) Find the *hwagentd_log_levels* parameter and enter the decimal number calculated using the instructions above.

c) Save the modified hwagentd.conf file.

d) Choose one of the following options to make the Hardware Management Agent reread the hwagentd.conf file.

On Linux and Solaris OSs, (Solaris OS: refresh) the Hardware Management Agent can be manually restarted, which forces the hwagentd.conf to be reread. Depending on the host OS the Hardware Management Agent is running on, restart the Hardware Management Agent.

On Windows OSs, you can restart the service using the Microsoft Management Console Services snap-in. The Hardware Management Agent rereads the hwagentd.conf file with the modified *hwagentd_log_levels* parameter.

2.9 Enhancing an OS with Utility Software

Utility software is a kind of system software designed to help analyze, configure, optimize and maintain the computer. A single piece of utility software is usually called *a utility (abbr. util) or tool.* Utility software should be contrasted with application software, which allows users to do things like creating text documents, playing games, listening to music, or surfing the web. Rather than providing these kinds of user-oriented or output-oriented functionality, utility software usually focuses on *how* the computer infrastructure (including the computer hardware, operating system, and application software and data storage) operates. Due to this focus, utilities are often rather technical and targeted at people with an advanced level of computer knowledge.

Most utilities are highly specialized and designed to perform only a single task or a small range of tasks. However, some utility suites combine several features in one piece of software. Most major operating systems come with several pre-installed utilities.

Application Software: An application software focuses on how computer infrastructure (including computer hardware, OS, & application software & data storage) operates. It allows the users to do things like creating text documents, playing games, listening to music or surfing the web. It offers user-oriented or output-oriented functionality.

Utility vs Application Software

- Utilities are technical & targeted at people with an advanced level of computer knowledge.
- Most utilities are highly specialized and designed to perform only a single task or a small range of tasks.

- Some utility suites combine several features in one piece of software.
- Most major OSs come with several pre-installed utilities.

Utility Software Categories

Disk storage utilities

- Disk defragmenters can detect computer files whose contents are broken across several locations on the hard disk, and move the fragments to one location to increase efficiency.
- Disk checkers can scan the contents of a hard disk to find files or areas that are corrupted in some way, or were not correctly saved, and eliminate them.
- ***Disk cleaners*** help to find files that are unnecessary to the computer operation or take up considerable amounts of space. They help the user to decide what to delete if the hard disk is full.
- ***Disk space analyzer*** offers visualization of disk space usage by getting the size for each folder (including subfolders) & files in a folder or drive. These show the distribution of the used space.
- ***Disk partitions*** divide an individual drive into multiple logical drives, each with its file system which can be mounted by the OS and treated as an individual drive.
- ***Backup utilities*** make a copy of all information stored on a disk and restore either the entire disk (e.g. in an event of disk failure) or selected files (e.g. in an event of accidental deletion).
- ***Disk compression utilities*** can transparently compress/uncompress the contents of a disk, increasing the capacity of the disk.
- ***File managers*** provide a convenient method of performing routine data management tasks, such as deleting, renaming, cataloging, uncataloging, moving, copying, merging, generating, and modifying data sets.
- *Archive utilities:* The archive utilities output a stream or a single file when provided with a directory or a set of files. Unlike archive suites, these usually do not include compression or encryption capabilities.
- *System profilers* provide detailed information about the software installed and hardware attached to the computer.
- *System monitors* for monitoring resources and performance in a computer system.
- *Anti-virus utilities* scan for computer viruses.
- *Hex editors* directly modify the text or data of a file.
- *Data compression utilities* output a shorter stream or a smaller file when provided with a stream or file.
- *Cryptographic utilities* encrypt and decrypt streams and files.
- *Launcher applications* provide a convenient access point for application software.
- *Registry cleaners* clean and optimize the Windows registry by removing old registry keys that are no longer in use.
- *Network utilities* analyze the computer's network connectivity, configure network settings, check data transfer or log events.
- *Screensavers* were desired to prevent phosphor burn-in on CRT and plasma computer monitors by blanking the screen or filling it with moving images or patterns when the computer is not in use. Contemporary screensavers are used primarily for entertainment or security.

Task:

How we manage hardware in Operating Systems?

What are the different categories of Utilities?

Summary

- The computer system can be divided into four components—hardware, operating system, the application program, and the user.
- An operating system is an interface between your computer and the outside world.
- Multiuser is a term that defines an operating system or application software that allows concurrent access by multiple users of a computer.
- A system call is a mechanism used by an application program to request services from the OS.
- The kernel is a computer program at the core of a computer's operating system that has complete control over everything in the system.
- The kernel is the portion of the operating system code that is always resident in memory and facilitates interactions between hardware and software components.
- Utilities are often rather technical and targeted at people with an advanced level of computer knowledge.

Keywords

Directory Access Permissions: A directory's access permissions help to control access to the files in it. These affect the overall ability to use files and subdirectories in the directory.

File Access Permissions: Access permissions on a file control what can be done to the file's contents.

Graphical User Interfaces: Most of the modern computer systems support graphical user interfaces (GUI), and often include them. In some computer systems, such as the original implementation of Mac OS, the GUI is integrated into the kernel.

Real-Time Operating System (RTOS): Real-time operating systems are used to control machinery, scientific instruments, and industrial systems.

System Calls: In computing, a system call is a mechanism used by an application program to request service from the operating system based on the monolithic kernel or to system servers on operating systems based on the microkernel structure.

The Root Menu: If you move the pointer onto the root window (the "desktop" behind the windows) and press the correct mouse button (usually the first or third button, depending on your setup), you should see the root menu.

The xterm Window: One of the most important windows is an xterm window. xterm makes a terminal emulator window with a UNIX login session inside, just like a miniature terminal.

Utility Software: Kind of system software designed to help analyze, configure, optimize and maintain the computer. A single piece of utility software is usually called a utility or tool.

Lab Exercise: Draw a flow chart for components of a computer system.

Self Assessment

1. Which of the following provides the interface to access the services of the operating system?

A. API

B. System call

C. Library
D. Assembly instruction

2. What is Microsoft window?
A. An operating system
B. A graphics program
C. A word processing application
D. A database program
3. What is the use of directory structure in the operating system?
A. The directory structure is used to solve the problem of the network connection in OS.
B. It is used to store folders and files hierarchically.
C. It is used to store the program in file format.
D. All of the these

4. Which of the following operating system runs on the server?
A. Batch OS
B. Real-time OS
C. Distributed OS
D. Network OS

5. The operating system work between
A. User and computer
B. Network and user
C. One user to another user
D. All of these

6. Which of the following operating systems supports only real-time applications?
A. Batch OS
B. Distributed OS
C. Real-time OS
D. Network OS

7. Which of the following is/are the functions of the operating system?
i. Sharing hardware among users
ii. Allowing users to share data among themselves
iii. Recovering from errors
iv. Scheduling resources among users
A. i, ii, and iii only
B. ii, iii, and iv only
C. i, iii, and iv only
D. All i, ii, iii, and iv

8. __________ is a large kernel containing virtually the complete operating system, including scheduling, file system, device drivers, and memory management.

A. Multilithic kernel
B. Monolithic kernel
C. Micro kernel
D. Macro kernel

9. ______________ is a kind of system software designed to help analyze, configure, optimize and maintain the computer.
A. Application software
B. Compiler
C. Kernel
D. Utility software

10. Which of the following offer(s) visualization of disk space usage by getting the size for each folder (including subfolders) and files in folder or drive.
A. Disk cleaner
B. Disk space analyser
C. Disk compression
D. Disk fragmentary

11. _______ makes a terminal emulator window with a UNIX login session inside, just like a miniature terminal.
A. xterm
B. DOS
C. Session terminal
D. Disk engine

12. ____________ focuses on how computer infrastructure (including computer hardware, OS, & application software & data storage) operates.
A. Kernel
B. Compiler
C. Application software
D. Utility software

13. ___________ make a copy of all information stored on a disk, and restore either the entire disk (e.g. in an event of disk failure) or selected files (e.g. in an event of accidental deletion).
A. Archive utilities
B. Backup utilities
C. Disk storage utilities
D. Fragmentation utilities

14. The access permissions on a file control what can be done to the file's contents are:
A. Disk access permissions
B. File access permissions
C. Backup access permissions
D. Directory access permissions

15. The ability of an OS to have more than one program (task) open at one time is called _______.

A. threading

B. programming

C. multitasking

D. utilization

Answers for Self Assessment

1.	B	2.	A	3.	B	4.	D	5.	A
6.	C	7.	D	8.	B	9.	D	10.	B
11.	A	12.	B	13.	C	14.	D	15.	C

Review Questions

1. What is an operating system? Give its types.
2. Define System Calls. Give their types also.
3. What are the different functions of an operating system?
4. What are user interfaces in the operating system?
5. Define GUI and Command-Line?
6. What is the setting of focus?
7. Define the xterm Window and Root Menu?
8. What is sharing of files? Also, give the commands for sharing the files?
9. Give steps of Managing hardware in Operating Systems.
10. What is the difference between Utility Software and Application software?
11. Define Real-Time Operating System (RTOS) and Distributed OS?
12. Describe how to run the program in the Operating system.

Further Readings

Introduction to Computers, by Peter Norton, Publisher: McGraw Hill, Sixth Edition

Web Links

http://computer.howstuffworks.com/operating-system.htm

What is an Operating System? Types of OS, Features and Examples (guru99.com)